

Stochastic thermodynamics

Work, heat, and rare trajectories

A trapped bead or a molecular coordinate fluctuates substantially from one trajectory to the next. Thermodynamics still applies, but its bookkeeping must distinguish trajectory-level random quantities from ensemble inequalities. An overdamped coordinate in one thermal bath is a useful starting point.

1 Trajectory thermodynamics and fluctuation relations

Let x move in a potential $U(x, \lambda)$ controlled by a prescribed protocol $\lambda(t)$, with an additional applied force $f(x, t)$. For constant friction γ and bath temperature T ,

$$\gamma \dot{x} = -\partial_x U + f + \sqrt{2\gamma k_B T} \xi(t), \quad \langle \xi(t) \xi(t') \rangle = \delta(t - t'). \quad (1)$$

The noise has zero mean. This model assumes separation from the bath's fast time scales and the Einstein relation for thermal noise; an active fluctuating force cannot simply be assigned the same temperature.

Define work done *on* the system and heat delivered *to* the bath by

$$\delta W = \partial_\lambda U d\lambda + f \circ dx, \quad \delta Q_{\text{bath}} = (-\partial_x U + f) \circ dx, \quad dU = \delta W - \delta Q_{\text{bath}}. \quad (2)$$

The circle denotes a Stratonovich integral. It permits the ordinary chain rule $dU = \partial_x U \circ dx + \partial_\lambda U d\lambda$. Even though the Langevin noise is additive, Itô and Stratonovich conventions differ for the heat integral because its integrand depends on x .

Entropy includes the probability of the state. Let $p(x, t)$ be the ensemble probability density. Up to a fixed reference measure, the stochastic system entropy is $s(x, t) = -k_B \ln p(x, t)$. For a trajectory of duration τ ,

$$\Delta s_{\text{tot}} = -k_B \ln \frac{p(x_\tau, \tau)}{p(x_0, 0)} + \frac{Q_{\text{bath}}}{T}. \quad (3)$$

The first term is essential when the distribution changes. Heat divided by temperature alone is not the total entropy production.

Under local detailed balance, the logarithm of the ratio between the forward path probability and the appropriately reversed path probability is $\Delta s_{\text{tot}}/k_B$. Here the reversed experiment uses the reversed protocol and starts in the forward final distribution; position is even under time reversal. Normalization of the reversed path measure gives the integral fluctuation theorem [1].

The second law from a fluctuation identity. For the regular diffusion above, with the required path-measure support,

$$\langle e^{-\Delta s_{\text{tot}}/k_B} \rangle = 1, \quad \langle \Delta s_{\text{tot}} \rangle \geq 0. \quad (4)$$

The inequality follows directly from Jensen's inequality. A single trajectory can have negative entropy production without contradicting it. The identity also implies, for any $a > 0$,

$$\Pr(\Delta s_{\text{tot}} \leq -a) \leq e^{-a/k_B}, \quad (5)$$

by applying Markov's inequality to the nonnegative variable $e^{-\Delta s_{\text{tot}}/k_B}$. Large negative fluctuations are constrained quantitatively, not forbidden.

For purely potential driving, $f = 0$, prepare the initial state in canonical equilibrium and change λ at any finite rate. The inclusive work obeys the Jarzynski equality [2]

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \beta = (k_B T)^{-1}, \quad \langle W \rangle \geq \Delta F. \quad (6)$$

Here ΔF compares the equilibrium free energies of the two endpoint potentials. The actual final distribution need not be equilibrated. This work relation and the total-entropy identity use different reversed initial ensembles; their exponents are not generally identical for a nonequilibrium final state.

An exactly checkable trap shift. Take $U(x, \lambda) = \kappa(x - \lambda)^2/2$, with $\kappa > 0$. Initially $\lambda = 0$ and x_0 is Gaussian with variance $k_B T/\kappa$. Shift the trap center instantaneously to a . Since the coordinate has no time to move,

$$W = \frac{\kappa}{2}(x_0 - a)^2 - \frac{\kappa x_0^2}{2} = \frac{\kappa a^2}{2} - \kappa a x_0. \quad (7)$$

Thus $\langle W \rangle = \kappa a^2/2$ and $\text{Var}(W) = \kappa a^2 k_B T$. The Gaussian moment-generating function gives

$$\langle e^{-\beta W} \rangle = \exp \left[-\beta \langle W \rangle + \frac{\beta^2}{2} \text{Var}(W) \right] = 1. \quad (8)$$

This agrees with $\Delta F = 0$: translating a harmonic trap leaves its partition function unchanged. There is no heat transfer during the instantaneous shift; dissipation occurs during subsequent relaxation. Rare negative-work realizations are needed for the exponential average to equal one.

Single-molecule pulling experiments motivate this distinction between work and free energy. In practice, exponential averages can converge slowly because rare trajectories carry large weight. Hidden coordinates, feedback, and chemical reservoirs require their own thermodynamic accounting before applying these formulas.

2 Probability currents and path reversal

Abbreviate the mechanical force as $F(x, t) = -\partial_x U + f$ and define the thermal diffusivity $D = k_B T/\gamma$. Here $F(x, t)$ denotes force, whereas ΔF in the work relation denotes a free-energy difference. The Langevin equation generates

$$\partial_t p = -\partial_x J, \quad J = \frac{F}{\gamma} p - D \partial_x p. \quad (9)$$

Assume periodic boundaries or sufficiently rapid decay so boundary contributions vanish. For the Shannon entropy $S = -k_B \int p \ln p dx$,

$$\dot{S} = -k_B \int (\partial_t p)(\ln p + 1) dx = -k_B \int J \partial_x \ln p dx. \quad (10)$$

The Stratonovich convention implies $\langle \dot{Q}_{\text{bath}} \rangle = \int F J dx$. Adding bath and system contributions therefore gives

$$\langle \dot{s}_{\text{tot}} \rangle = k_B \int J \left[\frac{F}{k_B T} - \partial_x \ln p \right] dx = \boxed{k_B \int \frac{J^2}{Dp} dx \geq 0}. \quad (11)$$

The final expression follows by dividing Eq. (9) by Dp . It identifies irreversible probability flow as the source of average entropy production. It is not obtained by replacing a random heat by its most probable value.

Equilibrium and a driven steady state are different. With $f = 0$ and a static potential, $p_{\text{eq}} \propto e^{-U/(k_B T)}$ makes $J = 0$. In contrast, take a ring of circumference L with a constant force F_0 . Then

$$p_{\text{st}} = \frac{1}{L}, \quad J_{\text{st}} = \frac{F_0}{\gamma L}, \quad \langle \dot{s}_{\text{tot}} \rangle = \frac{F_0^2}{\gamma T}. \quad (12)$$

The density is time independent, so $\dot{S} = 0$, but particles continue to wind around the ring and dissipate work. Winding must be retained when computing displacement and heat; reducing every displacement modulo L would erase the current.

For static conservative driving, define the nonequilibrium free-energy functional

$$\mathcal{F}[p] = \int p U dx + k_B T \int p \ln p dx. \quad (13)$$

Up to the fixed reference measure, $\mathcal{F}[p] - F_{\text{eq}} = k_B T \int p \ln(p/p_{\text{eq}}) dx$. Combining the first law with Eq. (11) gives

$$\frac{d\mathcal{F}}{dt} = -T \langle \dot{s}_{\text{tot}} \rangle \leq 0. \quad (14)$$

Relaxation consumes the excess free energy stored in the initial distribution. With an explicitly driven potential, an additional mean work-rate term appears; this monotonicity cannot be used unchanged.

Path reversal explains the exponential work average. For constant D , a midpoint discretization assigns a path weight containing the quadratic action $\int (\dot{x} - F/\gamma)^2 dt / (4D)$ and a drift-Jacobian term. On reversing the path and protocol, the velocity changes sign while the position-dependent force is evaluated at the same mirrored points. The Jacobian terms cancel in the ratio. The quadratic difference gives

$$\ln \frac{\mathcal{P}_F[x|x_0]}{\mathcal{P}_R[\tilde{x}|x_\tau]} = \frac{1}{k_B T} \int_0^\tau F \circ dx = \beta Q_{\text{bath}}. \quad (15)$$

Here $\tilde{x}(t) = x(\tau - t)$. Multiplying by $p_0(x_0)/p_\tau(x_\tau)$ yields the total-entropy relation in the overview. For the work theorem, instead choose canonical initial ensembles for both the forward and reverse experiments and set $f = 0$. Their initial-density ratio satisfies

$$\ln \frac{p_{\text{eq},0}(x_0)}{p_{\text{eq},\tau}(x_\tau)} = \beta [U(x_\tau, \lambda_\tau) - U(x_0, \lambda_0) - \Delta F]. \quad (16)$$

Using $W = \Delta U + Q_{\text{bath}}$ gives the forward/reverse path ratio $e^{\beta(W - \Delta F)}$. Group paths with the same work, noting that the reversed path has work $-W$, to obtain Crooks' relation [3]:

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta(W - \Delta F)}. \quad (17)$$

Multiplying by $e^{-\beta W}$ and integrating over W proves Jarzynski's equality by normalization of P_R . The crossing $P_F(W) = P_R(-W)$ occurs at $W = \Delta F$ when the distributions overlap there.

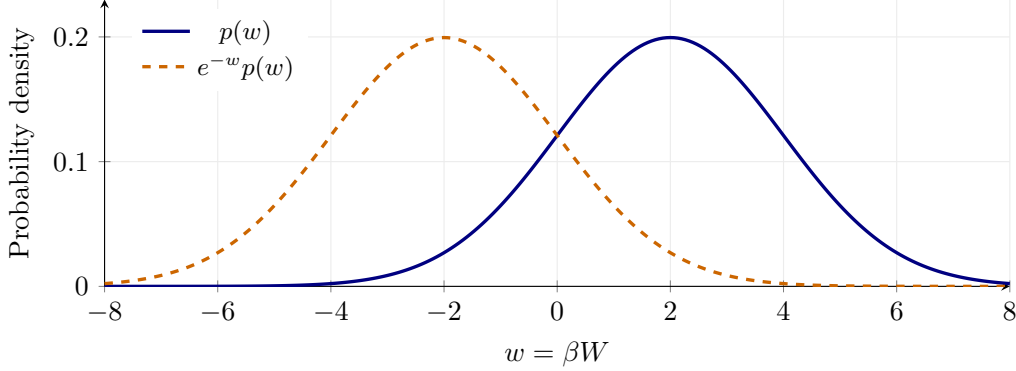


Figure 1: Trap-shift example with $\langle \beta W \rangle = 2$. The work density and its exponentially weighted density each integrate to one; the latter emphasizes negative-work events. Analytic Gaussian curves.

For this Gaussian example, $w = \beta W$ has variance 4. Multiplication by e^{-w} shifts its mean from +2 to -2 without changing the variance. This explains why a good estimate of ordinary mean work need not imply a good estimate of the exponential average.

3 Driven traps and biochemical networks

The instantaneous trap shift is one limit of a tractable finite-time protocol. Let $U = \kappa(x - Vt)^2/2$, where V is a constant trap speed, and start in equilibrium at $t = 0$. Define $\tau_r = \gamma/\kappa$. Averaging the Langevin equation gives

$$\frac{d\langle x \rangle}{dt} + \frac{\langle x \rangle}{\tau_r} = \frac{Vt}{\tau_r}, \quad \langle x(t) \rangle = Vt - V\tau_r(1 - e^{-t/\tau_r}). \quad (18)$$

The mean lag is initially small and approaches $V\tau_r$. Since the parameter work rate is $\dot{W} = -\kappa V(x - Vt)$, integration over a protocol of duration τ yields

$$\langle W \rangle = \gamma V^2 [\tau - \tau_r(1 - e^{-\tau/\tau_r})]. \quad (19)$$

The fluctuations $\delta x = x - \langle x \rangle$ obey an undriven Ornstein–Uhlenbeck equation. With the equilibrium initial variance,

$$\langle \delta x(t) \delta x(s) \rangle = \frac{k_B T}{\kappa} e^{-|t-s|/\tau_r}. \quad (20)$$

Work is a linear functional of this Gaussian process, so it is Gaussian. Integrating the covariance gives

$$\text{Var}(W) = \kappa V^2 k_B T \int_0^\tau dt \int_0^\tau ds e^{-|t-s|/\tau_r} \quad (21)$$

$$= 2\gamma V^2 k_B T [\tau - \tau_r(1 - e^{-\tau/\tau_r})] = 2k_B T \langle W \rangle. \quad (22)$$

Since $\Delta F = 0$, this variance–mean relation verifies Jarzynski’s equality exactly for every duration. It depends on the harmonic model and protocol; it is not a general Gaussian approximation for an arbitrary molecule.

For a fixed total displacement $a = V\tau$, put $r = \tau/\tau_r$. Then

$$\frac{\langle W \rangle}{\kappa a^2} = \frac{1}{r} - \frac{1 - e^{-r}}{r^2} \longrightarrow \begin{cases} 1/2, & r \rightarrow 0, \\ 1/r, & r \rightarrow \infty. \end{cases} \quad (23)$$

The fast limit recovers the sudden shift; the slow limit gives $\langle W \rangle \simeq \gamma a^2/\tau$. For example, $\kappa a^2 = 4k_B T$ gives $\beta \langle W \rangle = 2$ in the fast limit, $4/e \simeq 1.472$ at $r = 1$, and approximately 0.360 at $r = 10$.

At the end of dragging, the mean displacement from the new trap center stores excess energy $\kappa[V\tau_r(1 - e^{-r})]^2/2$. Relaxing afterward at fixed trap center adds no work but releases that stored energy as heat. Thus the division between heat dissipated during driving and during relaxation depends on where the observation interval ends.

From a coordinate to a biochemical network. For molecular motors and enzymes, a Markov jump model is often more practical than a resolved coordinate. Let p_i be the probability of internal state i and w_{ij} the transition rate from i to j . Then

$$\dot{p}_i = \sum_{j \neq i} (p_j w_{ji} - p_i w_{ij}). \quad (24)$$

Each resolved transition channel must be paired with its reverse. If several chemical mechanisms join the same states, keep the channel labels when computing dissipation. After including appropriate internal-state entropy and reservoir exchanges in local detailed balance, the total mean entropy-production rate is

$$\dot{S}_{\text{tot}} = k_B \sum_{i < j} (p_i w_{ij} - p_j w_{ji}) \ln \frac{p_i w_{ij}}{p_j w_{ji}} \geq 0. \quad (25)$$

Every summand is nonnegative because $(a - b) \ln(a/b) \geq 0$ for positive a, b . Stationarity only requires the currents into each state to balance; it does not require each edge current to vanish. A cycle current can persist.

A tightly coupled motor cycle. Suppose one forward cycle consumes one ATP and advances a distance ℓ against an opposing load F_{load} . Let $\Delta\mu > 0$ be the free energy released by ATP hydrolysis at the specified reservoir concentrations. Local detailed balance around a complete cycle gives its dimensionless affinity

$$\mathcal{A} = \ln \frac{\prod_{\text{cycle}} w_{ij}}{\prod_{\text{cycle}} w_{ji}} = \frac{\Delta\mu - F_{\text{load}}\ell}{k_B T}. \quad (26)$$

Internal state free-energy differences cancel around the cycle. If the net cycle current is J_{cyc} , then the entropy-production rate is $k_B J_{\text{cyc}} \mathcal{A}$, the mechanical power is $F_{\text{load}} \ell J_{\text{cyc}}$, and the chemical input power is $\Delta\mu J_{\text{cyc}}$.

For this unicyclic, tightly coupled model, stall occurs at $F_{\text{load}} \ell = \Delta\mu$. Illustrative values $\Delta\mu = 20k_B T$, $\ell = 8 \text{ nm}$, and $T = 300 \text{ K}$ give 10.4 pN. Slippage and futile cycles can reduce the stall force and dissipate ATP even at zero velocity.

What an experiment or trajectory analysis must specify. Specify temperature, work convention, initial preparation, and sampling rate. Establish whether the observed states admit Markov dynamics and identify unobserved chemical channels. Hidden currents can dissipate energy even when observed currents vanish; an equilibrium-looking coarse-grained trajectory does not establish equilibrium of the full motor.

References

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- [3] G. E. Crooks, “Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences,” *Phys. Rev. E* **60**, 2721 (1999). [arXiv:cond-mat/9901352](#).